**BE-101 · BTech First Year · UIT, Barkatullah University, Bhopal**

# Engineering Mathematics-I (Calculus & Linear Algebra) — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

**KAISE PADHEIN** Har unit mein pehle definition, phir formula, phir ek chhota solved example diya hai. Exam se pehle sirf **Formula** aur **Yaad Rakho** boxes revise kar lo, numericals ke steps example se copy karo.

## UNIT I Calculus — Evolutes, Integrals, Beta-Gamma, Volume & Surface

### 1. Evolutes and Involutes

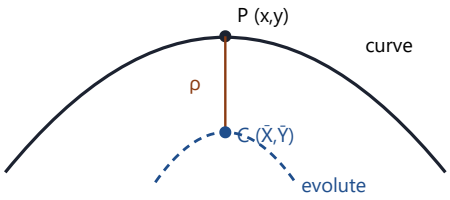
Kisi curve ke har point par ek **circle of curvature** hota hai jiska centre **centre of curvature** kehlata hai. Curve par point move karta hai to ye centre bhi ek nayi curve banata hai.

**DEFINITION** **Evolute** = locus of centre of curvature of a given curve. Agar curve C' curve C ka evolute hai, to C ko C' ka **Involute** kehte hain.

**FORMULA** $y_1=dy/dx$, $y_2=d^2y/dx^2$

$$\rho = (1+y_1^2)^{3/2} / |y_2|$$

$$\bar{X} = x - y_1(1+y_1^2)/y_2, \quad \bar{Y} = y + (1+y_1^2)/y_2$$




Normal PC = ρ; C ka locus = evolute

#### Steps to find Evolute

1. $y_1$, $y_2$ nikalo (parametric ho to t ke terms mein).
2. $\bar{X}$, $\bar{Y}$ formula mein daalo.
3. Parameter (x, y ya t) ko eliminate karo → $\bar{X}$, $\bar{Y}$ ke beech relation = evolute.

#### Properties

- Curve ka **normal** evolute ka **tangent** hota hai.
- Do points ke radii of curvature ka difference = evolute ke corresponding arc ki length.
- Ek curve ka evolute unique hai, par involutes infinite hote hain (string unwinding — gear teeth profile involute hota hai).

**SOLVED EXAMPLE** Parabola $y^2 = 4ax$ ka evolute.
$x = at^2$, $y = 2at \Rightarrow y_1 = 1/t$, $y_2 = -1/(2at^3)$, $1+y_1^2 = (t^2+1)/t^2$.
$\bar{X} = at^2 + 2a(t^2+1) = 3at^2 + 2a$, $\bar{Y} = 2at - 2at(t^2+1) = -2at^3$.
$(\bar{X}-2a)^3 = 27a^3t^6$, $\bar{Y}^2 = 4a^2t^6 \Rightarrow$ **$27a\bar{Y}^2 = 4(\bar{X}-2a)^3$** (semi-cubical parabola).

**YAAD RAKHO** Ellipse $x^2/a^2+y^2/b^2=1$ ka evolute: **$(ax)^{2/3} + (by)^{2/3} = (a^2-b^2)^{2/3}$**. Astroid $x^{2/3}+y^{2/3}=a^{2/3}$ ka evolute: $(x+y)^{2/3}+(x-y)^{2/3} = 2a^{2/3}$.

### 2. Evaluation of Definite and Improper Integrals

#### Definite integral properties

| Property | Result |
|---|---|
| P1 | $\int_a^b f(x)dx = \int_a^b f(t)dt$ ; $\int_a^b f = -\int_b^a f$ |
| P2 (King's rule) | $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ ; $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$ |
| P3 | $\int_{-a}^a f(x)dx = 2\int_0^a f(x)dx$ if f even ; = 0 if f odd |
| P4 | $\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx$ if $f(2a-x)=f(x)$ ; = 0 if $f(2a-x)=-f(x)$ |
| Wallis | $\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \frac{(n-1)(n-3)\ldots}{n(n-2)\ldots} \times$ ($\pi/2$ if n even ; 1 if n odd) |

**EXAMPLE** $I = \int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}\,dx$. P2 lagao ($x \to \pi/2-x$): $I = \int \sqrt{\cos x}/(\sqrt{\cos x}+\sqrt{\sin x})\,dx$. Dono add: $2I = \int_0^{\pi/2} 1\,dx = \pi/2 \Rightarrow$ **$I = \pi/4$**.

### Improper integrals

Jab limit infinite ho (**Kind I**: $\int_a^{\infty}$) ya integrand interval mein infinite ho jaye (**Kind II**: $\int_0^1 dx/\sqrt{x}$) — tab integral ko limit ke roop mein evaluate karte hain: $\int_a^{\infty} f\,dx = \lim_{t\to\infty} \int_a^t f\,dx$. Limit finite → **convergent**, warna **divergent**.

**STANDARD** $\int_1^{\infty} dx/x^p$ : convergent if $p > 1$, divergent if $p \le 1$.
$\int_0^1 dx/x^p$ : convergent if $p < 1$, divergent if $p \ge 1$.

**EXAMPLE** $\int_0^{\infty} e^{-x}dx = \lim_{t\to\infty}(1-e^{-t}) = 1$ (conv.). $\int_1^{\infty} dx/x = \lim \log t = \infty$ (div.).

## 3. Beta and Gamma Functions

**BETA** $B(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx, \quad m, n > 0$

**GAMMA** $\Gamma(n) = \int_0^{\infty} e^{-x}\,x^{n-1}\,dx, \quad n > 0$

### Properties (sab yaad karne hain)

1. $\Gamma(n+1) = n\Gamma(n)$ ; $\Gamma(n+1) = n!$ ($n$ positive integer)
2. $\Gamma(1) = 1$, **$\Gamma(1/2) = \sqrt{\pi}$**, $\Gamma(3/2) = \sqrt{\pi}/2$
3. $B(m,n) = B(n,m)$ (symmetric)
4. **$B(m,n) = \Gamma(m)\Gamma(n) / \Gamma(m+n)$**
5. $B(m,n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta\,d\theta$
6. $B(m,n) = \int_0^{\infty} x^{m-1}/(1+x)^{m+n}\,dx$
7. $\Gamma(n)\Gamma(1-n) = \pi / \sin n\pi, \quad 0<n<1$
8. Duplication: $\Gamma(m)\Gamma(m+\tfrac{1}{2}) = \sqrt{\pi}\,\Gamma(2m) / 2^{2m-1}$

**MOST USED**

$$\int_0^{\pi/2} \sin^p\theta \cos^q\theta\,d\theta = \frac{\Gamma((p+1)/2)\,\Gamma((q+1)/2)}{2\,\Gamma((p+q+2)/2)}, \quad p, q > -1$$

**EXAMPLE 1** $\int_0^{\infty} e^{-x^2}dx$. Put $x^2 = t$, $dx = dt/(2\sqrt{t})$:
$= \tfrac{1}{2}\int_0^{\infty} e^{-t}t^{-1/2}dt = \tfrac{1}{2}\Gamma(\tfrac{1}{2}) =$ **$\sqrt{\pi}/2$**.

**EXAMPLE 2** $\int_0^{\pi/2} \sin^4\theta \cos^2\theta\,d\theta = \Gamma(5/2)\Gamma(3/2)/(2\Gamma(4))$
$= (3\sqrt{\pi}/4)(\sqrt{\pi}/2)/(2\cdot 6) =$ **$\pi/32$**.

**COMMON MISTAKE** $\Gamma(n) = (n-1)!$, $n!$ nahi. $\Gamma(5) = 4! = 24$. Aur $\Gamma(0)$, $\Gamma(-1)$ defined nahi hain.

## 4. Applications: Volume and Surface Area of Revolution

Ek plane curve ko kisi axis ke around ghumane par **solid of revolution** banta hai. Thin disc (radius $y$, thickness $dx$) ka volume $\pi y^2 dx$; thin ring ka surface $2\pi y \cdot ds$.

| Curve form | Volume (about x-axis) | Surface Area (about x-axis) |
|---|---|---|
| Cartesian $y=f(x)$ | $V = \pi\int_a^b y^2\,dx$ | $S = 2\pi\int_a^b y\,\sqrt{1+(dy/dx)^2}\,dx$ |
| About y-axis $x=g(y)$ | $V = \pi\int_c^d x^2\,dy$ | $S = 2\pi\int_c^d x\,\sqrt{1+(dx/dy)^2}\,dy$ |
| Parametric $x(t)$, $y(t)$ | $V = \pi\int y^2\,(dx/dt)\,dt$ | $S = 2\pi\int y\,\sqrt{\dot{x}^2+\dot{y}^2}\,dt$ |
| Polar $r=f(\theta)$ (initial line) | $V = (2\pi/3)\int r^3\sin\theta\,d\theta$ | $S = 2\pi\int r\sin\theta\,\sqrt{r^2+(dr/d\theta)^2}\,d\theta$ |

**EXAMPLE** Circle $x^2+y^2=a^2$ ko x-axis ke around ghumao (sphere).
$V = \pi\int_{-a}^{a}(a^2-x^2)dx = \pi[a^2x - x^3/3]_{-a}^{a} =$ **$4\pi a^3/3$**.
$y = \sqrt{(a^2-x^2)}$, $dy/dx = -x/y \Rightarrow y\sqrt{(1+x^2/y^2)} = \sqrt{(y^2+x^2)} = a$. $S = 2\pi\int_{-a}^{a} a\,dx =$ **$4\pi a^2$**.

**YAAD RAKHO** Revolution x-axis ke around → y ka formula ($y^2$, $y\,ds$). y-axis ke around → x ka formula. Symmetric curve ho to 2×(half) le lo.

# UNIT II Calculus — Mean Value Theorems, Expansions, Limits, Extrema

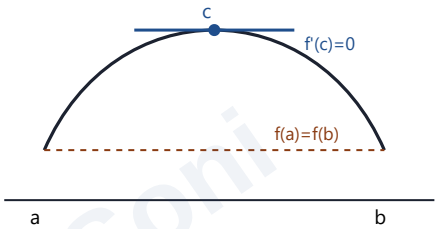
## 1. Rolle's Theorem

**STATEMENT** Agar $f(x)$ (i) **continuous** hai $[a,b]$ par, (ii) **differentiable** hai $(a,b)$ mein, aur (iii) **$f(a) = f(b)$**, to kam se kam ek $c \in (a,b)$ exist karega jahan **$f'(c) = 0$**.

**Geometrical meaning:** curve par kam se kam ek point aisa hai jahan tangent x-axis ke parallel hai.

**EXAMPLE** $f(x) = x^2 - 4x + 3$ on [1,3]. Polynomial ⇒ continuous & differentiable. $f(1) = 0 = f(3)$. $f'(c) = 2c - 4 = 0 \Rightarrow \mathbf{c = 2 \in (1,3)}$. Verified.

**FAIL CASE** $f(x)=|x|$ on $[-1,1]$: $f(-1)=f(1)$ par $x=0$ par differentiable nahi ⇒ Rolle's apply nahi hota.



Rolle's: horizontal tangent at c

## 2. Mean Value Theorems

### (a) Lagrange's Mean Value Theorem (LMVT)

Agar f continuous [a,b] par aur differentiable (a,b) mein, to ∃ c ∈ (a,b):

**LMVT**

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \text{ya} \quad f(a+h) = f(a) + h\,f'(a+\theta h),\ 0<\theta<1$$

**Meaning:** chord AB ke parallel kam se kam ek tangent hai. Rolle's = LMVT ka special case (f(a)=f(b)).

**EXAMPLE** $f(x) = x^3$ on [0,2]: $f'(c) = 3c^2 = (8-0)/(2-0) = 4 \Rightarrow c = 2/\sqrt{3} \approx 1.155 \in (0,2)$. Verified.

### (b) Cauchy's Mean Value Theorem

f, g dono continuous [a,b] par, differentiable (a,b) mein, aur $g'(x) \neq 0$, to ∃ c ∈ (a,b):

**CMVT**

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}$$

**EXAMPLE** $f = e^x$, $g = e^{-x}$ on [a,b]. LHS $= (e^b - e^a)/(e^{-b} - e^{-a}) = -e^{a+b}$. RHS $= e^c/(-e^{-c}) = -e^{2c}$. ⇒ **c = (a+b)/2** (A.M. of a, b). g(x)=x lene par CMVT → LMVT ban jata hai.

## 3. Taylor's and Maclaurin's Theorems

**TAYLOR'S THEOREM**

$$f(a+h) = f(a) + h\,f'(a) + \frac{h^2}{2!}f''(a) + \ldots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n$$

Lagrange's remainder: $R_n = \frac{h^n}{n!}f^{(n)}(a+\theta h)$; Cauchy's remainder: $R_n = \frac{h^n(1-\theta)^{n-1}}{(n-1)!}f^{(n)}(a+\theta h)$, $0<\theta<1$.

Powers of (x−a) mein: $f(x) = f(a) + (x-a)f'(a) + (x-a)^2/2!\,f''(a) + \ldots$

**Maclaurin's theorem:** Taylor mein a = 0, h = x: $f(x) = f(0) + x\,f'(0) + x^2/2!\,f''(0) + x^3/3!\,f'''(0) + \ldots$

| Function | Maclaurin expansion | Function | Maclaurin expansion |
|---|---|---|---|
| $e^x$ | $1 + x + x^2/2! + x^3/3! + \ldots$ | $\log(1+x)$ | $x - x^2/2 + x^3/3 - \ldots$ $(-1<x\le 1)$ |
| $\sin x$ | $x - x^3/3! + x^5/5! - \ldots$ | $(1+x)^n$ | $1 + nx + n(n-1)x^2/2! + \ldots$ |
| $\cos x$ | $1 - x^2/2! + x^4/4! - \ldots$ | $\tan x$ | $x + x^3/3 + 2x^5/15 + \ldots$ |
| $\sinh x$ | $x + x^3/3! + x^5/5! + \ldots$ | $\cosh x$ | $1 + x^2/2! + x^4/4! + \ldots$ |

**EXAMPLE (TAYLOR)** sin x ko powers of $(x - \pi/2)$ mein: $a = \pi/2$, $f(a)=1$, $f'(a)=\cos(\pi/2)=0$, $f''(a)=-1$, $f'''(a)=0$, $f^{iv}(a)=1$.
$\sin x = 1 - (x-\pi/2)^2/2! + (x-\pi/2)^4/4! - \ldots$

**EXAMPLE (MACLAURIN)** $e^x \sin x = (1+x+x^2/2+x^3/6+\ldots)(x-x^3/6+\ldots)$
$= x + x^2 + (1/2 - 1/6)x^3 + \ldots = \mathbf{x + x^2 + x^3/3 + \ldots}$

## 4. Indeterminate Forms and L'Hospital's Rule

Seven indeterminate forms: **0/0, ∞/∞, 0×∞, ∞−∞, $0^0$, $1^\infty$, $\infty^0$**.

**L'HOSPITAL'S RULE** Agar lim f(x)/g(x) form 0/0 ya ∞/∞ ho, to $\lim_{x\to a} f(x)/g(x) = \lim_{x\to a} f'(x)/g'(x)$ (numerator aur denominator ko **alag-alag** differentiate karo, quotient rule nahi). Form bana rahe to dobara lagao.

| Form | Conversion trick | Solved example |
|---|---|---|
| 0/0 | Direct L'Hospital | $\lim_{x\to0}(\sin x - x)/x^3 = \lim(\cos x - 1)/3x^2 = \lim(-\sin x)/6x =$ **−1/6** |
| 0×∞ | f·g = f/(1/g) | $\lim_{x\to0^+} x \log x = \lim (\log x)/(1/x) = \lim (1/x)/(-1/x^2) = \lim(-x) =$ **0** |
| ∞−∞ | LCM le kar 0/0 banao | $\lim_{x\to0}(1/x - 1/\sin x) = \lim(\sin x - x)/(x \sin x) \to (\cos x-1)/(\sin x + x\cos x) \to -\sin x/(2\cos x - x \sin x) =$ **0** |
| $1^\infty$, $0^0$, $\infty^0$ | $y = f^g$, log y = g·log f | $\lim_{x\to0}(1+x)^{1/x}$: log y = log(1+x)/x → 1/(1+x) → 1 ⇒ y = **e** |

**COMMON MISTAKE** Har step par pehle check karo ki form abhi bhi 0/0 ya ∞/∞ hai. Agar nahi, to L'Hospital mat lagao — direct value daalo.

## 5. Maxima and Minima

### (a) One variable

1. f'(x) = 0 solve karo → stationary points.
2. f''(x) < 0 ⇒ **maximum**; f''(x) > 0 ⇒ **minimum**.
3. f'' = 0 ho to higher derivative: pehla non-zero derivative **even order** ka ho to extremum (negative → max), **odd** ho to neither (point of inflexion).

**EXAMPLE** $f = x^3 - 6x^2 + 9x + 1$. $f' = 3(x-1)(x-3) = 0 \Rightarrow x = 1, 3$. $f'' = 6x - 12$.
x=1: f''=−6 < 0 ⇒ max, f(1) = **5**.
x=3: f''=6 > 0 ⇒ min, f(3) = **1**.

### (b) Two variables f(x,y)

1. $f_x = 0$, $f_y = 0$ solve → stationary points.
2. $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$ nikalo.

| Condition | Result |
|---|---|
| $rt - s^2 > 0$, r < 0 | Maximum |
| $rt - s^2 > 0$, r > 0 | Minimum |
| $rt - s^2 < 0$ | Saddle point (no extremum) |
| $rt - s^2 = 0$ | Doubtful — further study |

**EXAMPLE** $f = x^3 + y^3 - 3axy$ (a > 0). $f_x = 3x^2 - 3ay = 0$, $f_y = 3y^2 - 3ax = 0$ ⇒ points (0,0), (a,a). r = 6x, s = −3a, t = 6y.
(a,a): $rt - s^2 = 36a^2 - 9a^2 = 27a^2 > 0$, r = 6a > 0 ⇒ **minimum**, $f_{min} = -a^3$. (0,0): $rt - s^2 = -9a^2 < 0$ ⇒ **saddle point**.

**LAGRANGE'S MULTIPLIERS** Constraint φ(x,y,z)=0 ke saath f extremise karna ho: F = f + λφ banao, $F_x=F_y=F_z=0$ aur φ=0 solve karo. (Jaise: x+y+z = a fixed ho to xyz max at x=y=z=a/3.)

# UNIT III Sequences, Series and Fourier Series

## 1. Convergence of Sequence

**DEFINITION** Sequence $\{a_n\}$ **convergent** hai agar $\lim_{n\to\infty} a_n = l$ (finite, unique). Limit ±∞ ho to **divergent**; alag-alag values ke beech jhoole to **oscillatory**.

- {1/n} → 0 : convergent
- $\{(1+1/n)^n\}$ → e : convergent
- $\{n^2\}$ → ∞ : divergent
- $\{(-1)^n\}$ : oscillates finitely
- **Monotonic + bounded** ⇒ convergent
- Convergent ⇒ bounded (converse false)

## 2. Convergence of Series and Tests

Series $\Sigma u_n$ ke partial sum $S_n = u_1+...+u_n$. Agar lim $S_n$ finite ⇒ series **convergent**.

**NECESSARY CONDITION** $\Sigma u_n$ convergent ⇒ lim $u_n$ = 0. Converse false: Σ1/n mein $u_n \to 0$ par series diverge karti hai. Isliye lim $u_n \neq 0$ ⇒ seedha **divergent**.

**GEOMETRIC** $\Sigma r^n$: convergent if |r| < 1 (sum a/(1−r)); divergent if r ≥ 1; oscillatory if r ≤ −1.

**P-SERIES** $\Sigma 1/n^p$: convergent if **p > 1**, divergent if p ≤ 1. (Comparison ke liye sabse useful.)

| Test (positive terms) | Statement | Kab use karein |
|---|---|---|
| Comparison (limit form) | lim $u_n/v_n$ = finite non-zero ⇒ $\Sigma u_n$, $\Sigma v_n$ dono saath conv./div. | $u_n$ algebraic (polynomial/root) ho; $v_n = 1/n^p$ (highest powers ka ratio) |
| D'Alembert's Ratio | lim $u_{n+1}/u_n$ = L: L<1 conv., L>1 div., L=1 fail | n!, $a^n$, $x^n$ wale terms |
| Raabe's | lim $n(u_n/u_{n+1} - 1)$ = k: k>1 conv., k<1 div. | Ratio test fail (L = 1) |
| Logarithmic | lim $n\cdot\log(u_n/u_{n+1})$ = k: k>1 conv., k<1 div. | Ratio fail aur e ya $n^n$ aaye |
| Cauchy's Root | lim $(u_n)^{1/n}$ = L: L<1 conv., L>1 div. | $u_n = (\ldots)^n$ type |
| Cauchy's Integral | f(x) positive decreasing: $\Sigma f(n)$ aur $\int_1^\infty f(x)dx$ saath conv./div. | 1/(n log n) type |
| Leibnitz (alternating) | $\Sigma(-1)^{n-1}u_n$: $u_n$ decreasing & $u_n \to 0$ ⇒ convergent | Alternating signs |

**Absolute convergence:** $\Sigma|u_n|$ convergent ⇒ $\Sigma u_n$ absolutely convergent. $\Sigma u_n$ conv. par $\Sigma|u_n|$ div. ⇒ **conditionally convergent** (e.g., 1 − ½ + ⅓ − …).

**NOTE** Grewal/Indian books mein ratio test ulta likha hota hai: lim $u_n/u_{n+1}$ = k, k>1 conv., k<1 div. Dono same baat hai — sirf ratio ulta hai.

**EXAMPLE 1** $\Sigma n^2/2^n$: $u_{n+1}/u_n = (n+1)^2/(2n^2) \to 1/2 < 1$ ⇒ **convergent**.
$\Sigma 1/(n^2+1)$: $v_n = 1/n^2$, ratio → 1, $\Sigma 1/n^2$ conv. (p=2) ⇒ **convergent**.

**EXAMPLE 2** $\Sigma (1+1/n)^{-n^2}$: $(u_n)^{1/n} = (1+1/n)^{-n} \to 1/e < 1$ ⇒ **convergent** (root test).
$\Sigma \sqrt{n}/(n+1)$: ~ $1/\sqrt{n}$, p=½ ⇒ **divergent**.

## 3. Power Series and Taylor's Series

**Power series:** $\Sigma a_n x^n = a_0 + a_1x + a_2x^2 + \ldots$ . Ye |x| < R ke liye (absolutely) convergent, |x| > R ke liye divergent; x = ±R par alag se check karo.

**RADIUS OF CONVERGENCE**

$$R = \lim_{n\to\infty} |a_n / a_{n+1}| \quad \text{ya} \quad 1/R = \lim |a_n|^{1/n}$$

**EXAMPLE** $\Sigma x^n/n$: $a_n/a_{n+1} = (n+1)/n \to 1 \Rightarrow R = 1$. x = 1: $\Sigma 1/n$ divergent. x = −1: $\Sigma(-1)^n/n$ convergent (Leibnitz). ⇒ Interval of convergence **[−1, 1)**.

**Taylor's series:** agar f ke sabhi derivatives exist karein aur remainder $R_n \to 0$, to $f(x) = \Sigma_{n=0}^{\infty} f^{(n)}(a)(x-a)^n/n!$ (a = 0 par Maclaurin's series). Ye ek power series hai in (x−a). Example: $e^x$ ki series sab x ke liye converge karti hai (R = ∞); log(1+x) ki R = 1.

## 4. Fourier Series

Periodic function ko sine aur cosine terms ke infinite sum mein likhna. **Dirichlet conditions:** f periodic, single-valued, finite; finite number of discontinuities aur maxima/minima ek period mein. Discontinuity par series = $[f(x^-)+f(x^+)]/2$.

**INTERVAL (−Π, Π) OR (0, 2Π)**

$$f(x) = a_0/2 + \Sigma_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$$

$a_0 = (1/\pi)\int f(x)dx$, $a_n = (1/\pi)\int f(x)\cos nx\, dx$, $b_n = (1/\pi)\int f(x)\sin nx\, dx$ (integration over one period of length 2π). Interval (−l, l) ho to nx → nπx/l aur 1/π → 1/l.

**EVEN / ODD IN (−Π,Π)** f **even** ⇒ $b_n = 0$ (sirf cosine). f **odd** ⇒ $a_0 = a_n = 0$ (sirf sine).
Useful: sin nπ = 0, cos nπ = $(-1)^n$.

**EXAMPLE** f(x) = x in (−π,π) (odd): $b_n = (2/\pi)\int_0^{\pi} x \sin nx\, dx = 2(-1)^{n+1}/n$
⇒ x = 2(sin x − sin2x/2 + sin3x/3 − …)

### Half Range Sine and Cosine Series

Jab f(x) sirf (0, π) [ya (0, l)] mein diya ho, to use odd extension karke **sine series** ya even extension karke **cosine series** mein expand karte hain.

| Series | Interval (0, π) | Interval (0, l) |
|---|---|---|
| Half range **sine**: f = Σ $b_n$ sin(·) | $b_n = (2/\pi)\int_0^{\pi} f(x) \sin nx\, dx$ | $b_n = (2/l)\int_0^{l} f(x) \sin(n\pi x/l)\, dx$ |
| Half range **cosine**: f = $a_0/2$ + Σ $a_n$ cos(·) | $a_0 = (2/\pi)\int_0^{\pi} f\, dx$, $a_n = (2/\pi)\int_0^{\pi} f \cos nx\, dx$ | $a_n = (2/l)\int_0^{l} f(x) \cos(n\pi x/l)\, dx$ |

**SINE SERIES OF F = 1 IN (0,Π)** $b_n = (2/\pi)\int_0^{\pi} \sin nx\, dx$ = $2(1-(-1)^n)/(n\pi)$ = $4/(n\pi)$ for odd n, 0 for even n. ⇒ $1 = (4/\pi)(\sin x + \sin 3x/3 + \sin 5x/5 + \ldots)$

**COSINE SERIES OF F = X IN (0,Π)** $a_0 = \pi$; $a_n = (2/\pi)\cdot((-1)^n-1)/n^2 = -4/(\pi n^2)$ (odd n), 0 (even). ⇒ $x = \pi/2 - (4/\pi)(\cos x + \cos 3x/3^2 + \cos 5x/5^2 + \ldots)$. x = 0 rakho ⇒ **$1 + 1/3^2 + 1/5^2 + \ldots = \pi^2/8$**.

### Parseval's Theorem

**PARSEVAL'S IDENTITY** $(-\pi, \pi)$ par: $(1/\pi)\int_{-\pi}^{\pi} [f(x)]^2\, dx = a_0^2/2 + \Sigma(a_n^2 + b_n^2)$

Half range (0, l): sine series ⇒ $(2/l)\int_0^{l} f^2\, dx = \Sigma b_n^2$ ; cosine series ⇒ $(2/l)\int_0^{l} f^2\, dx = a_0^2/2 + \Sigma a_n^2$.

**EXAMPLE** $x = 2\Sigma(-1)^{n+1} \sin nx/n$ in $(-\pi,\pi)$, $b_n^2 = 4/n^2$. LHS = $(1/\pi)\int_{-\pi}^{\pi} x^2 dx = (1/\pi)(2\pi^3/3) = 2\pi^2/3$. ⇒ $2\pi^2/3 = 4\Sigma 1/n^2$ ⇒ **$\Sigma\, 1/n^2 = \pi^2/6$**.

## UNIT IV Vector Spaces

### 1. Vector Space

**DEFINITION** Non-empty set V, field F (R ya C) ke upar **vector space** hai agar do operations — vector addition (u+v) aur scalar multiplication (αu) — define hon aur ye axioms satisfy karein (∀ u,v,w ∈ V, α,β ∈ F):

A1. Closure: u + v ∈ V

A2. Commutative: u + v = v + u

A3. Associative: (u+v)+w = u+(v+w)

A4. Zero vector 0: u + 0 = u

A5. Inverse −u: u + (−u) = 0

M1. Closure: αu ∈ V

M2. α(u+v) = αu + αv

M3. (α+β)u = αu + βu

M4. (αβ)u = α(βu)

M5. 1·u = u

**Examples:** $R^n$ (n-tuples), $P_n$ (degree ≤ n polynomials), $M_{m\times n}$ (all m×n matrices), C[a,b] (continuous functions). **Non-example:** set of polynomials of exactly degree 2 (sum ka degree gir sakta hai → closure fail).

### 2. Vector Sub Space

**DEFINITION** V ka non-empty subset W **subspace** hai agar W khud usi addition & scalar multiplication ke saath vector space ho.

**ONE-STEP TEST** W subspace ⇔ **αu + βv ∈ W** for all u, v ∈ W, α, β ∈ F. (Shortcut: 0 ∈ W hona hi chahiye — agar nahi to subspace nahi.)

**YES** W = {(x, y, 0)} ⊂ $R^3$ (xy-plane): $\alpha(x_1,y_1,0)+\beta(x_2,y_2,0)$ = (..., ..., 0) ∈ W ✓.
W = {(x,y,z): x+y+z = 0} bhi subspace hai.

**NO** W = {(x, y): x + y = 1}: (0,0) ∉ W ⇒ subspace nahi.
W = {(x,y): x ≥ 0}: (−1)(1,0) ∉ W ⇒ nahi.

Facts: do subspaces ka **intersection** hamesha subspace; **union** generally nahi (sirf jab ek dusre ke andar ho). {0} aur V khud trivial subspaces hain.

### 3. Linear Combination of Vectors

$v = \alpha_1 v_1 + \alpha_2 v_2 + \ldots + \alpha_n v_n$ ($\alpha_i$ scalars) ko $v_1,\ldots,v_n$ ka **linear combination** kehte hain. Saare possible linear combinations ka set = **span** L(S), jo hamesha ek subspace hota hai.

**EXAMPLE** (3, 5) ko (1,1) aur (1,2) ke LC mein likho: a(1,1)+b(1,2) = (3,5) ⇒ a + b = 3, a + 2b = 5 ⇒ b = 2, a = 1. ∴ (3,5) = 1·(1,1) + 2·(1,2). (Equations inconsistent hon to LC possible nahi.)

### 4. Linearly Dependent and Linearly Independent Vectors

**DEFINITION** $\alpha_1 v_1 + \ldots + \alpha_n v_n = 0$ ka **sirf trivial solution** (sab $\alpha_i = 0$) ho ⇒ **Linearly Independent (LI)**. Koi non-zero $\alpha_i$ se bhi zero ban jaye ⇒ **Linearly Dependent (LD)** (yaani koi vector baaki ka LC hai).

**EXAMPLE (LD)** (1,2,3), (2,3,4), (3,4,5):
det = 1(15−16) − 2(10−12) + 3(8−9) = −1+4−3 = 0 ⇒ **LD**. Relation: $v_1 + v_3 = 2v_2$.

**EXAMPLE (LI)** (1,0,0), (1,1,0), (1,1,1): det of matrix = 1 ≠ 0 ⇒ **LI**.

**YAAD RAKHO** (i) n vectors $R^n$ mein: det ≠ 0 ⇒ LI, det = 0 ⇒ LD. Generally: rank of matrix of vectors = number of vectors ⇒ LI. (ii) Zero vector wala set hamesha LD. (iii) Do vectors LD ⇔ ek dusre ka scalar multiple. (iv) $R^n$ mein n se zyada vectors hamesha LD.

### 5. Basis of a Vector Space

**DEFINITION** Subset B ⊂ V **basis** hai agar (i) B **linearly independent** ho, aur (ii) B V ko **span** kare. Basis mein vectors ki sankhya = **dimension** (dim V).

| Space | Standard basis | Dimension |
|---|---|---|
| $R^n$ | $e_1$=(1,0,...,0), ..., $e_n$=(0,...,0,1) | n |
| $P_n$ | {1, x, $x^2$, ..., $x^n$} | n + 1 |
| $M_{m\times n}$ | $E_{ij}$ (sirf (i,j) entry 1) | mn |

n-dimensional space mein koi bhi n LI vectors automatically basis bana dete hain. Har vector basis vectors ka **unique** LC hota hai. Example: {(1,0,0),(1,1,0),(1,1,1)} $R^3$ ka basis hai (3 LI vectors, dim $R^3$ = 3).

### 6. Linear Transformations

**DEFINITION** T : U → V **linear transformation** hai agar (i) T(u + v) = T(u) + T(v), (ii) T(αu) = αT(u). Combined: **T(αu + βv) = αT(u) + βT(v)**.

- T(0) = 0 hamesha (check karne ka shortcut).
- **Kernel** (null space): ker T = {u : T(u) = 0}, dim = nullity.
- **Range**: R(T) = {T(u)}, dim = rank.
- Har LT T: $R^n$ → $R^m$ ko m×n matrix A se likh sakte hain: T(X) = AX.

**RANK-NULLITY THEOREM**

$$\text{rank}(T) + \text{nullity}(T) = \dim U$$

**EXAMPLE 1** T(x,y) = (x+y, x−y): T(αu+βv) = αT(u)+βT(v) check ⇒ **linear**.
T(x,y) = (x+1, y): T(0,0) = (1,0) ≠ 0 ⇒ **not linear**.

**EXAMPLE 2** T(x,y,z) = (x+y, y+z). ker: x+y=0, y+z=0 ⇒ (x,y,z) = k(1,−1,1) ⇒ nullity = 1. Range = $R^2$ ⇒ rank = 2. Check: 2 + 1 = 3 = dim $R^3$ ✓.

## UNIT V Matrices

### 1. Rank of a Matrix

**DEFINITION** Matrix A ka **rank ρ(A) = r** agar kam se kam ek r-order minor non-zero ho aur (r+1)-order ke saare minors zero hon. Rank elementary transformations se change nahi hota.

**Elementary transformations:** (i) $R_i \leftrightarrow R_j$ (ii) $R_i \to kR_i$, $k \neq 0$ (iii) $R_i \to R_i + kR_j$ (same for columns).

**Echelon form:** har non-zero row ka pehla non-zero element pichhli row se right mein; zero rows neeche. **Rank = number of non-zero rows.**

**Normal form:** $[I_r\ 0;\ 0\ 0]$ — yahan r = rank. PAQ = N form bhi isi se aata hai.

**EXAMPLE** A = [1 2 3; 2 4 6; 1 1 1].
$R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$: [1 2 3; 0 0 0; 0 −1 −2].
$R_2 \leftrightarrow R_3$: [1 2 3; 0 −1 −2; 0 0 0] ⇒ **ρ(A) = 2**.

## 2. Solution of Simultaneous Linear Equations by Elementary Transformation

System AX = B ke liye **augmented matrix [A | B]** banao, sirf **row** transformations se echelon (Gauss elimination) form mein lao, phir back-substitution karo. (Gauss-Jordan mein [A|B] → [I|X] tak le jaate hain.)

**EXAMPLE** x+y+z = 6, x+2y+3z = 14, x+4y+7z = 30.
[1 1 1 | 6; 1 2 3 | 14; 1 4 7 | 30] → $R_2-R_1$, $R_3-R_1$ → [1 1 1 | 6; 0 1 2 | 8; 0 3 6 | 24] → $R_3-3R_2$ → [1 1 1 | 6; 0 1 2 | 8; 0 0 0 | 0].
ρ(A) = ρ(A|B) = 2 < 3 ⇒ infinite solutions. z = k ⇒ y = 8 − 2k, x = 6 − y − z = **k − 2**. Solution: (k−2, 8−2k, k).

## 3. Consistency of Equations

| System | Condition (n = number of unknowns) | Result |
|---|---|---|
| Non-homogeneous AX = B | ρ(A) ≠ ρ(A\|B) | **Inconsistent** — no solution |
| | ρ(A) = ρ(A\|B) = n | Consistent, **unique** solution |
| | ρ(A) = ρ(A\|B) = r < n | Consistent, **infinite** solutions (n − r arbitrary constants) |
| Homogeneous AX = 0 (always consistent) | ρ(A) = n (\|A\| ≠ 0) | Only **trivial** solution X = 0 |
| | ρ(A) < n (\|A\| = 0) | **Non-trivial** (infinite) solutions |

**EXAM FAVOURITE** x+y+z = 6, x+2y+3z = 10, x+2y+λz = μ. Reduce: [1 1 1|6; 0 1 2|4; 0 0 λ−3|μ−10].
(i) **λ ≠ 3** ⇒ unique solution; (ii) **λ = 3, μ ≠ 10** ⇒ no solution; (iii) **λ = 3, μ = 10** ⇒ infinite solutions.

## 4. Eigen Values and Eigen Vectors

**DEFINITION** Square matrix A ke liye agar non-zero vector X aur scalar λ aise hon ki **AX = λX**, to λ = **eigen value** (characteristic root) aur X = **eigen vector**. Eigen values **characteristic equation |A − λI| = 0** se milte hain; har λ ke liye (A − λI)X = 0 solve karke X.

**3×3 SHORTCUT**

$$\lambda^3 - (\text{trace } A)\,\lambda^2 + (\text{sum of principal minors of order } 2)\,\lambda - |A| = 0$$

2×2 ke liye: $\lambda^2$ − (trace)λ + |A| = 0.

**EXAMPLE 2×2** A = [2 1; 1 2]: $\lambda^2 - 4\lambda + 3 = 0 \Rightarrow \lambda = 1, 3$.
λ = 3: [−1 1; 1 −1]X = 0 ⇒ x = y ⇒ X = $(1, 1)^T$.
λ = 1: [1 1; 1 1]X = 0 ⇒ x = −y ⇒ X = $(1, -1)^T$.

**EXAMPLE 3×3** A = [1 1 3; 1 5 1; 3 1 1]: trace = 7, minors sum = 4 − 8 + 4 = 0, |A| = −36.
$\lambda^3 - 7\lambda^2 + 36 = 0 \Rightarrow \lambda = -2, 3, 6$.
Eigen vectors: (−1,0,1), (1,−1,1), (1,2,1).

### Properties of eigen values

1. Sum of eigen values = **trace** (sum of diagonal)
2. Product of eigen values = **|A|**
3. Triangular/diagonal matrix: eigen values = diagonal elements
4. A aur $A^T$ ke eigen values same
5. $A^{-1}$ ke eigen values 1/λ; $A^k$ ke $\lambda^k$; kA ke kλ
6. Real symmetric matrix ke eigen values real; eigen vectors orthogonal
7. Koi eigen value 0 ⇔ |A| = 0 (singular)
8. Distinct eigen values ke eigen vectors LI

## 5. Cayley-Hamilton Theorem and Inverse

**STATEMENT** **Har square matrix apni characteristic equation ko satisfy karti hai.** Agar $|A - \lambda I| = \lambda^n + c_1\lambda^{n-1} + \ldots + c_n = 0$, to $A^n + c_1A^{n-1} + \ldots + c_nI = 0$.

**INVERSE** Dono side $A^{-1}$ se multiply ($c_n = \pm|A| \neq 0$):

$$A^{-1} = -(1/c_n)\,[A^{n-1} + c_1A^{n-2} + \ldots + c_{n-1}I]$$

Applications: $A^{-1}$ nikalna, higher powers ($A^3$, $A^4$ ...) ko lower powers mein likhna.

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**EXAMPLE 2×2** A = [1 2; 3 4]: $\lambda^2 - 5\lambda - 2 = 0 \Rightarrow A^2 - 5A - 2I = 0 \Rightarrow A^{-1} = (A - 5I)/2 = \frac{1}{2}[-4\ 2; 3\ -1] = [-2\ 1; 3/2\ -1/2]$.

**EXAMPLE 3×3** A = [2 −1 1; −1 2 −1; 1 −1 2]. trace = 6, principal minors = 3+3+3 = 9, |A| = 4.
Char. eq.: $\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0 \Rightarrow$ (C-H) $A^3 - 6A^2 + 9A - 4I = 0 \Rightarrow$ **$A^{-1} = (A^2 - 6A + 9I)/4$**.
$A^2 = [6\ -5\ 5; -5\ 6\ -5; 5\ -5\ 6] \Rightarrow A^2 - 6A + 9I = [3\ 1\ -1; 1\ 3\ 1; -1\ 1\ 3] \Rightarrow$ **$A^{-1} = \frac{1}{4}[3\ 1\ -1; 1\ 3\ 1; -1\ 1\ 3]$**.
Verify: $A \cdot A^{-1}$ ki row1·col1 = (6 − 1 − 1)/4 = 1 ✓. C-H verify karne ke liye pehle $A^2$, $A^3$ nikalkar equation mein daal kar zero matrix dikhao.

**YAAD RAKHO** Signs: 3×3 mein $\lambda^3 - S\lambda^2 + P\lambda - |A|$ (S = trace, P = minors ka sum). C-H se $A^{-1}$ mein last constant (|A|) se divide hota hai — |A| = 0 ho to inverse exist nahi karta.

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